PS 200C: Causal inference for social science

Causal Graphs and Causal Structural Models

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Outline

1. Introduction
   - Backdoor Criterion
   - Adjustment Criterion
2. DAG Revelations
3. Complaints and responses
4. Other ID from DAGs: Front-doors & napkins

Resources for this material

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- *Primer* (Pearl, Glymour, Jewell), Chapters 1.4, 1.5, 2, 3.
- Pearl, J. (2009). *Causality*

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- Some online materials:
  - *Causal Diagrams: Draw Your Assumptions Before Your Conclusions* course by Miguel Hernan on EdX
  - *A Crash Course in Causality: Inferring Causal Effects from Observational Data* course by Jason Roy on Coursera.
  - Richard McElreath lecture on "Science before statistics: causal inference". `https://www.youtube.com/watch?v=KNPYUVmY3NM&t=8565s`

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- Important things revealed by DAGs

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We need a language that allows us to (i) express our causal assumptions, (ii) make causal queries, (iii) determine if those queries can be answered; (iv) tell us how to get those answers.

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1. To do so we can use structural causal models and their graphical representations as directed acyclic graphs
2. Potential outcomes/counterfactuals are hypothetical realizations from these structure, telling us outcomes under particular interventions.
3. Tensions between approaches once raged but have mostly subsided; in (my view of) reality these traditions are all of a piece and have different strengths and weaknesses.

Graphs and Stuctural Causal Models

Structural Causal Models (SCMs) are formal ways of specifying a set of causal relationships. They consist of:

- a group of observed variables
- a group of unobserved exogenous variables, $\{U\}$
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E.g.,

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\begin{aligned}
X &= f_x(U_x) \\
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Y &= f_y(X, A, U_y) \\
U_y &\perp\!\!\!\perp U_x, \;\; U_y \perp\!\!\!\perp U_a
\end{aligned}
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where "exogenous" ancestors, $U = \{U_x, U_y, U_z\}$ with $U \sim p(u)$.

Life in an SCM

Note that:

- $\{f_x, f_a, f_y\}$ are (but do not have to be) non-parametric functions
- they are "invariant" or "modular", don't depend on how you got there
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- Population: The distribution $p(U)$ implies a distribution $p(observables)$.
- Individuals: If you knew all the values of U for a given unit and the details of every f, you would know the exact values of every variable.
- The invariance notion allow us write down what would happen (to the distributions or specific realizations) under counterfactuals that modify the model or set values in it.

Defining Causality

In this context causality means (*Primer*):

> *A variable X is a cause of a variable Y if Y in any way relies on X for its value... think of causation as a form of listening; X is a cause of Y if Y listens to X and decides its value in response to what it hears.*

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Still, specific causal estimands generally invoke potential outcomes, which are effectively tools to summarize causal relationships in this view.

SCM to Graph

For a model M such as:

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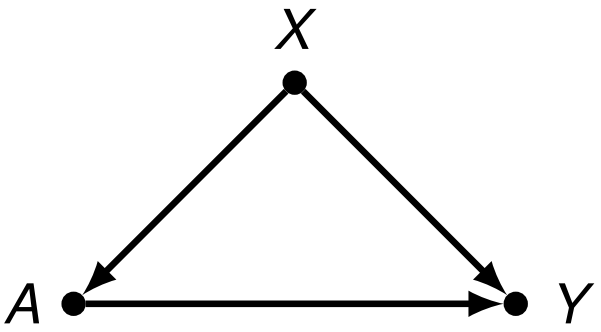

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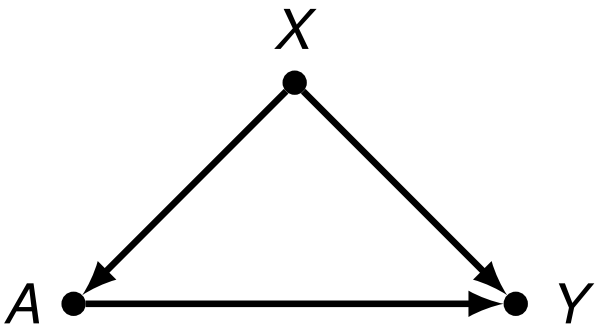

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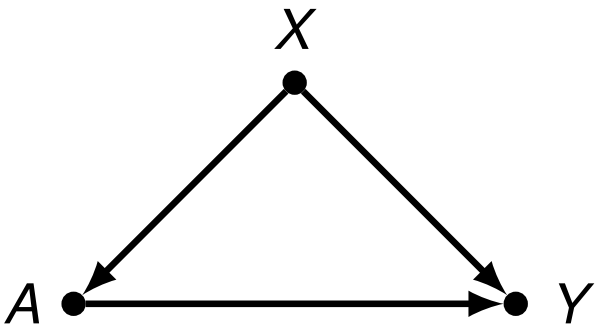

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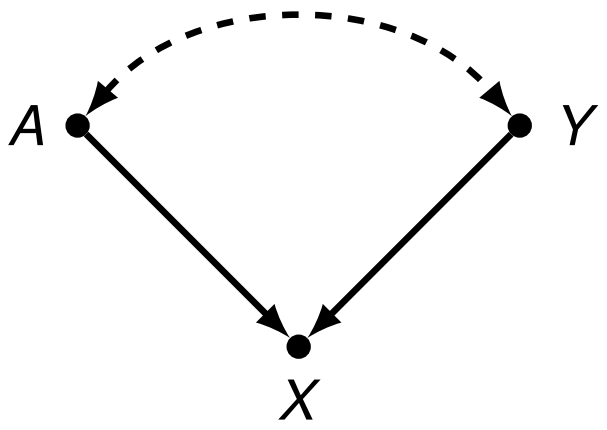
- You could show U_j pointing to each j, but usually omit these.

Missing edges, bidirected arcs

A missing edge means the value of A is not an argument into Y. *Its the missing edges that matter for the arguments we will need to make.*



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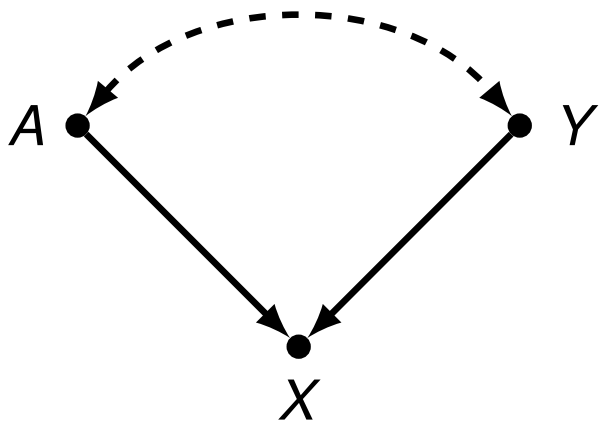

Some authors show unobserved common causes as a bidirected arc.

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5. descendants, children: effects of a given cause
6. paths: any series of connected nodes, regardless of direction
7. **d**irected **a**cyclic **g**raph: a graph where all edges are directed and you can never follow directed paths from a node back to itself.

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Beware colliders! Beware descendants of colliders!

Colliding is real

Pearl's favorite:

A: It rains

B: Sprinkler recently on

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Academic tenure:

A: Productivity

B: Originality

C: Tenure[1]

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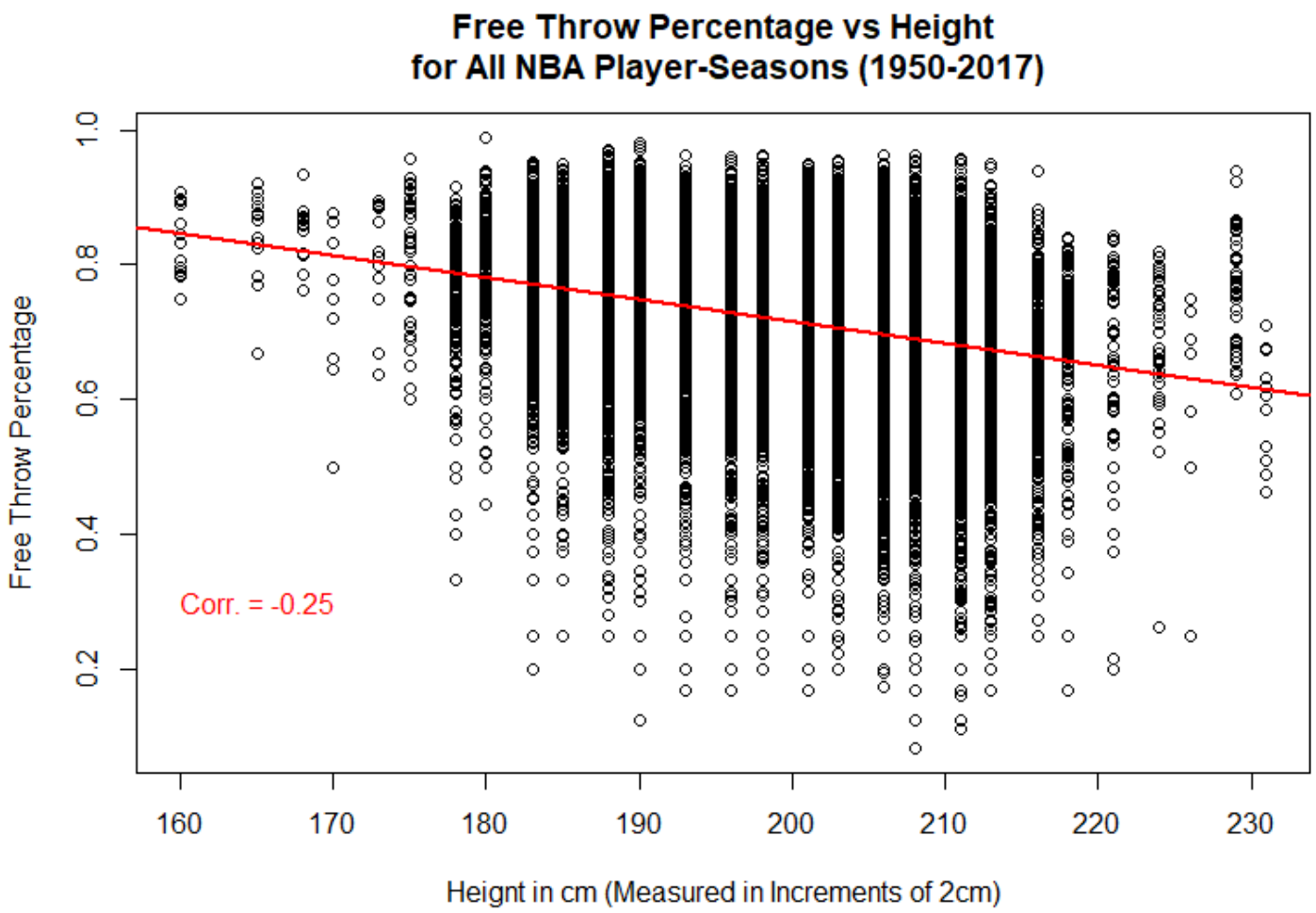

*Source: Adam Rohde, via `https://www.basketball-reference.com/` and `https://www.kaggle.com/datasets/drgilermo/nba-players-stats`.

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A and C are d-separated by a set of nodes Z if and only if:

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- *A and C are connected by a collider ($A \rightarrow B \leftarrow C$) s.t. the collision node is not in Z, and no descendant of it is in Z.*

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If nodes A and C are d-separated by Z, we expect $A \perp\!\!\!\perp B \mid Z$ in any dataset generated by that graph. The opposite is usually, but not always, true.

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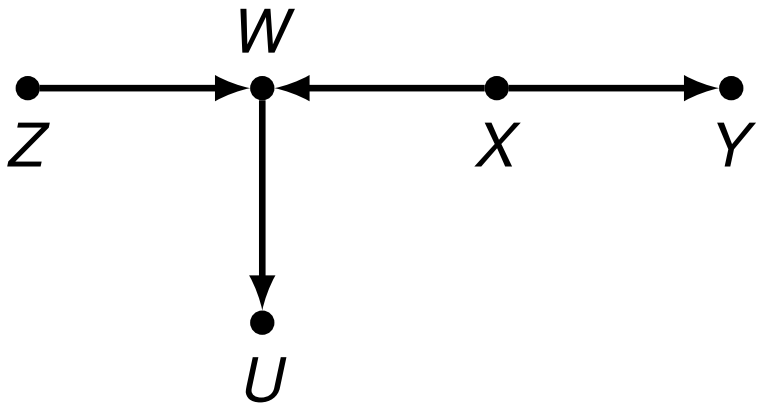
We will soon use conditional independence on DAGs to support claims of conditional ignorability, but not yet.

Practice



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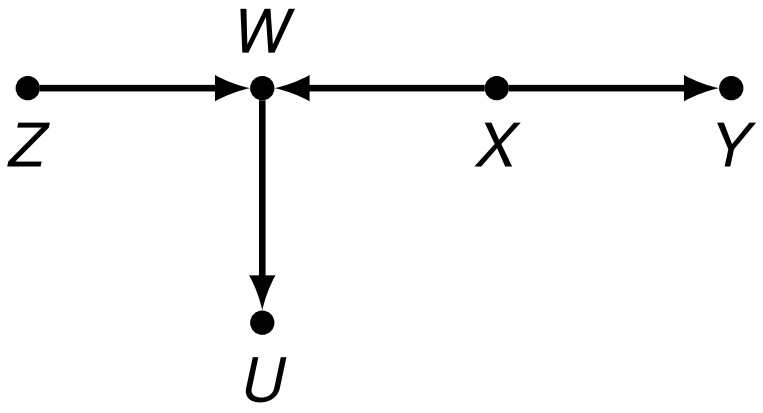

Looking at Z and Y:

- Are they d-separated conditional on $\{\}$?
- Are they d-separated conditional on W?
- Are they d-separated conditional on $\{W, X\}$?
- Are they d-separated conditional on U?

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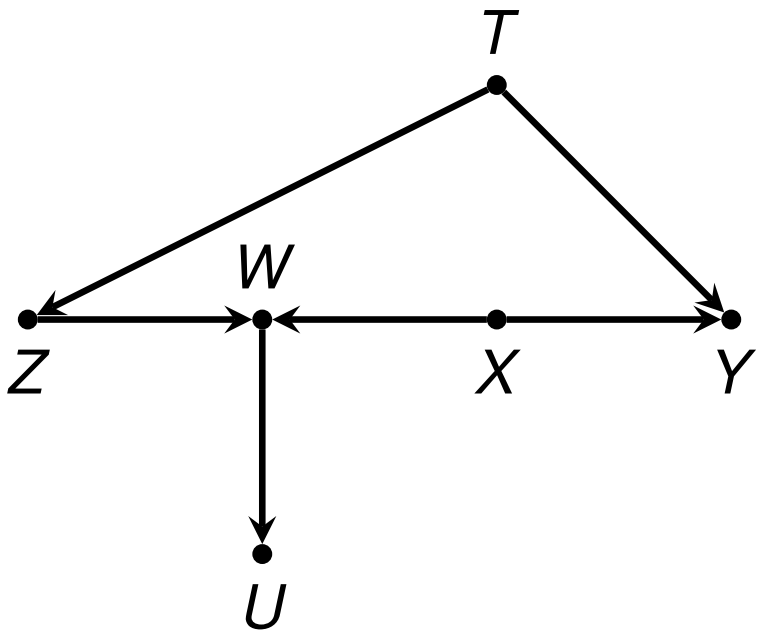

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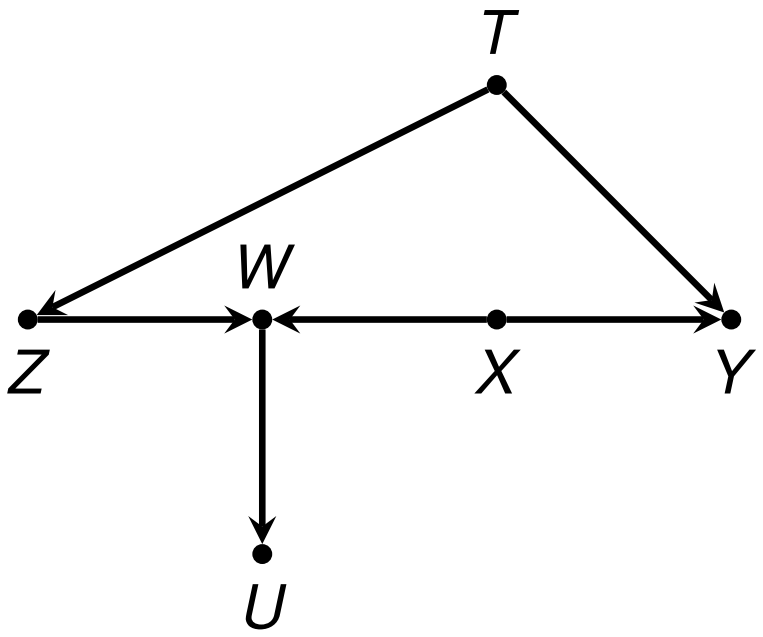

What sets d-separate Z from Y?

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- You can render them independent (*d*-separated) if,
  - they are independent already
  - you condition on common cause or intermediate variable
  - you don't condition on collider (or descendant thereof)

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- Backdoor paths between D and Y point into D.
  - These are typically unwanted sources of association we will want to remove from what we measure.

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This is 93+% of what people use DAGs for.

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- In PO land: "you have shown SOO/CI, so condition on Z", i.e. estimate treatment effect within strata of Z, average over strata.

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Once you find a conditioning set Z that satisfies the backdoor criterion relative to D on Y, you can

- In DAG land: compute $p(Y|do(d))$ using the adjustment formula,

$$p(Y|do(d)) = \sum_z p(Y|d,z)p(z)$$

  - This is very general; you are more used to asking for the ATE, which you can get from this:

$$\begin{aligned}\mathbb{E}[Y_1 - Y_0] &= \sum_z y * p(Y = y|d = 1, z)p(z) - \sum_z y * p(Y = y|d = 0, z \\ &= \sum_z (\mathbb{E}[Y|d = 1, z] - \mathbb{E}[Y|d = 0, z])p(z) \\ &= \text{our usual approach under SOO}\end{aligned}$$

- In PO land: "you have shown SOO/CI, so condition on Z", i.e. estimate treatment effect within strata of Z, average over strata.
- In practice: use your favorite technology to "adjust for" Z.

Example 1

$$\begin{tikzcd} SES \arrow[r, dashed] \arrow[d, dashed] & BP \arrow[d] \\ Drug \arrow[r] & Live \end{tikzcd}$$

where *SES* is socioeconomic status (unobserved), *BP* is blood pressure, *Drug* is a treatment, and *Live* is the outcome.

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- Go for backdoor criterion: what do you choose?

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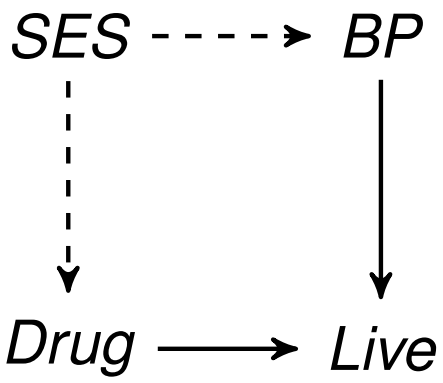

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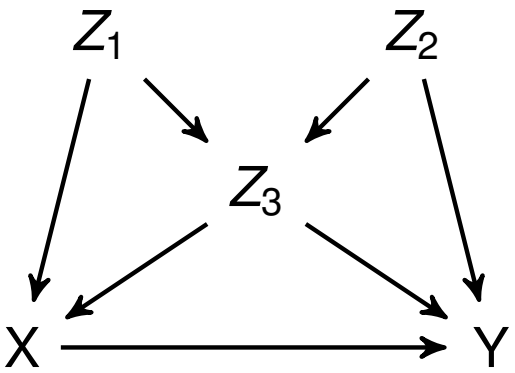

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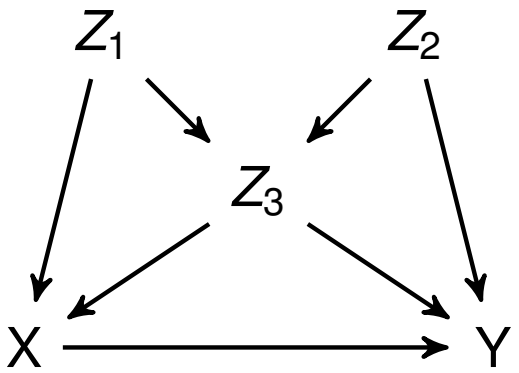

- What are the backdoor paths from X to Y?

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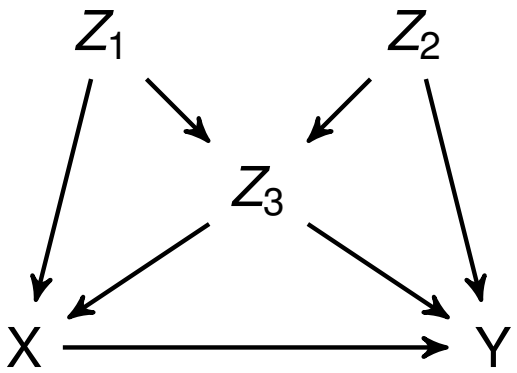
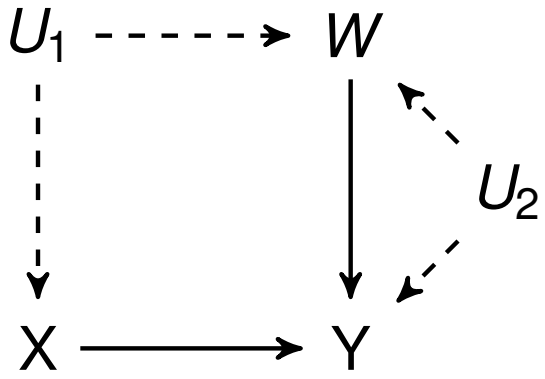


- What are the backdoor paths from X to Y?
- What sets block them?

How about here?



Adjustment Criterion

Shpitser, VanderWeele, Robins (2012)

[2] Also relevant: this paper defines descendants of X to include X itself.

Adjustment Criterion

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Noting that $G_{\overline{X}}$ is the graph representing $do(X = x)$[2]

Rule

Adjustment criterion. *Z satisfies the adjustment criterion relative to (X, Y) in G if (i) No element in Z is a descendant in $G_{\overline{X}}$ of any $W \notin X$ which lies on a proper causal path from X to Y; (ii) All non-causal paths in G from X to Y are blocked by Z.*

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Basically backdoor criterion, but allows conditioning on descendants of X when it is not on path to Y.

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Adjustment criterion vs. Backdoor

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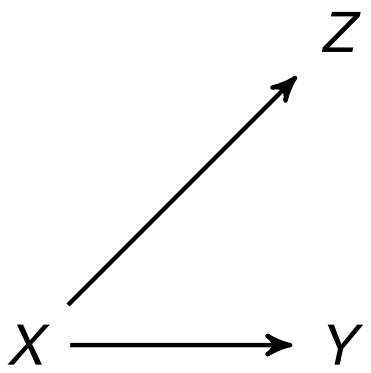

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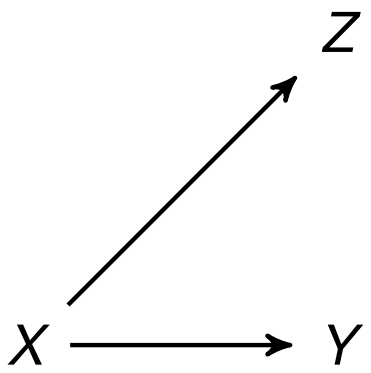

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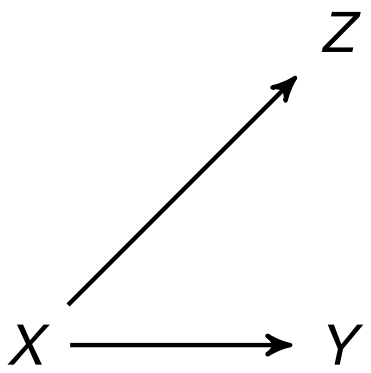

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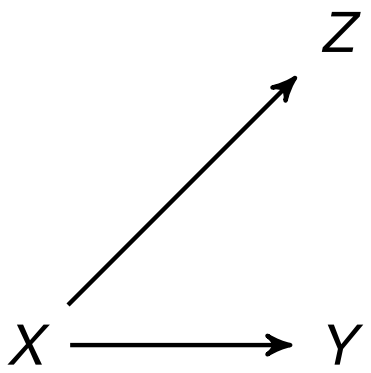

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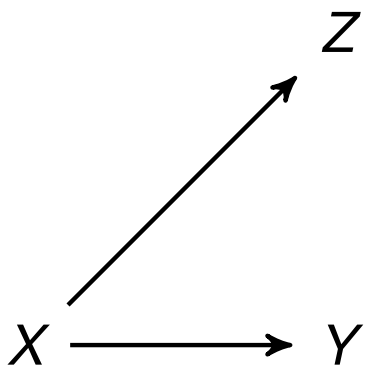

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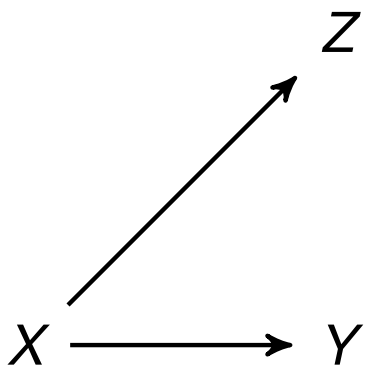

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- I don't know when this is useful in practice, but it is complete.
- Conditioning on such a Z could also weaken estimation, since conditionally on Z, variation in X would be lessened.

DAGs, POs, and conditional ignorability

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If some $\{Z\}$ satisfies backdoor/adjustment criterion for Y-D, then $Y_d \perp\!\!\!\perp D \mid \{Z\}$ (conditional ignorability)

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- Mostly people use DAGs just to check if conditioning on some $\{Z\}$ is a good idea, or to choose the $\{Z\}$ to condition on.
- DAGs and POs are compatible; DAGs help find failures in conditional ignorability claims. They are a thinking aid.

Outline

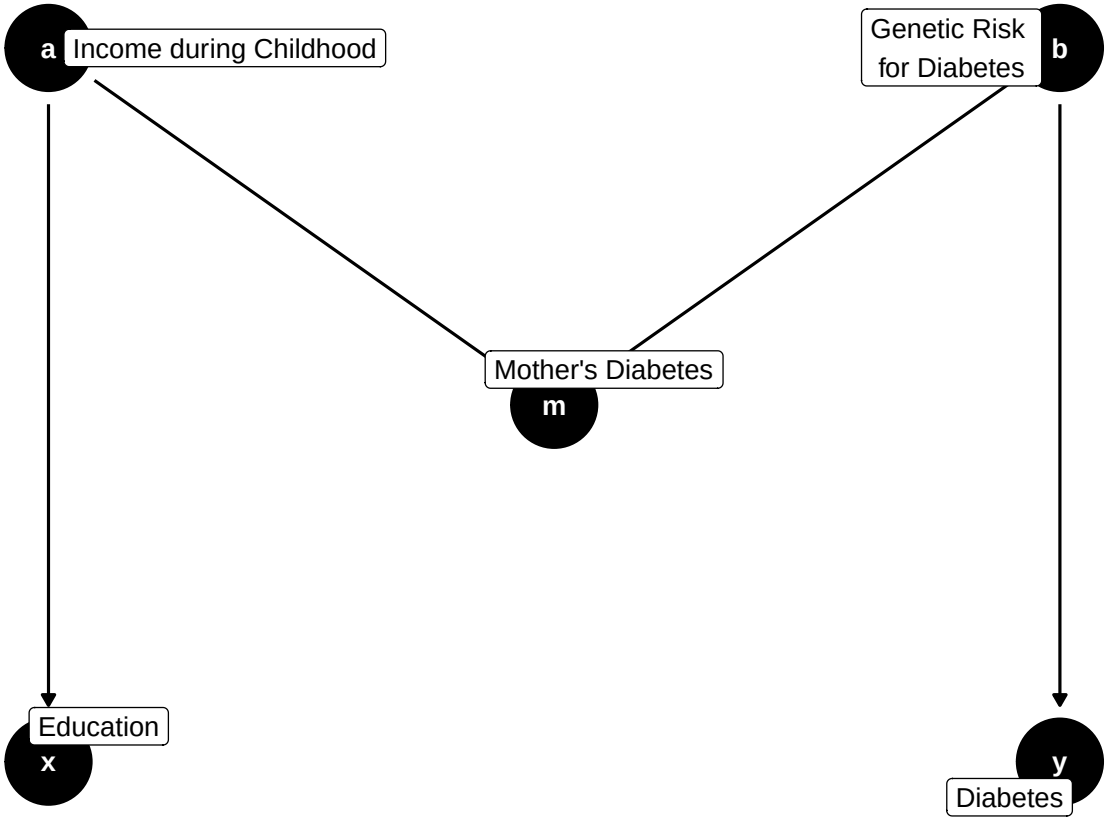
1. Introduction
   - Backdoor Criterion
   - Adjustment Criterion
2. DAG Revelations
3. Complaints and responses
4. Other ID from DAGs: Front-doors & napkins

Why bother? Some errors and revelations made clear by DAGs

- M-bias, butterfly bias, sample selection
- Stratifying on post-treatment variables
- Why you cannot control for Z and check D-Y for exclusion restriction.

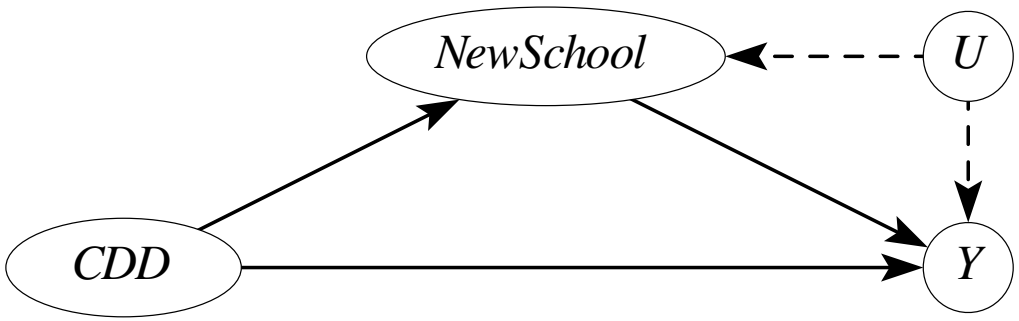
M-bias, butterly bias, sample selection

You already saw this but let's do another example[3]

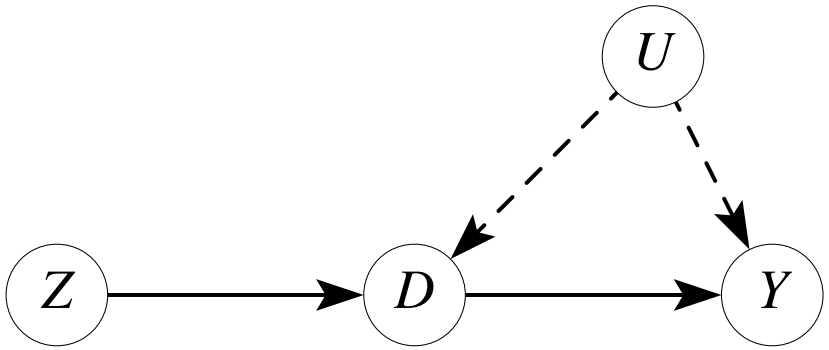


[3] Example and ggdag code courtesy of Malcolm Barrett, using an example borrowed from "Modern Epidemiology"

Stratifying on post-treatment trouble



You can't show exclusion by checking $Y \sim Z|D$



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- Given colliders and such, this enables you to find challenges to CI that are not obvious

2. Many ID strategies (IV, RDD, DID) depend on assumptions not on a DAG, and therefore cannot be "proven" or fully understood from DAG perspective.

- Good reason to be bilingual and not *just* rely on DAGs.
- But these approaches still require assumptions that can be shown on DAG *plus* some others, so you can still surface problems from the DAG side
- In fact DAGs are great for explaining some things, e.g. IV and the exclusion restriction, even if that's not enough.

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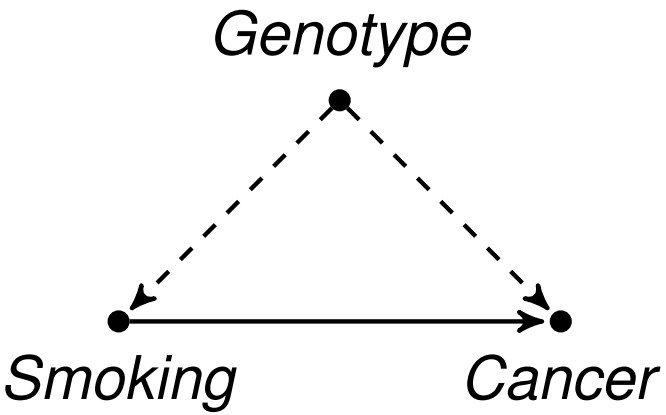

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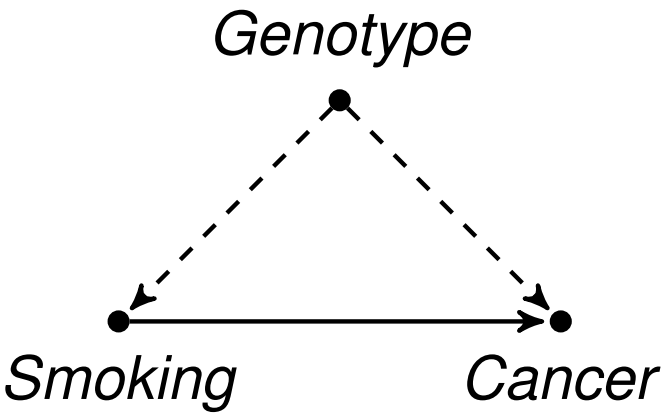

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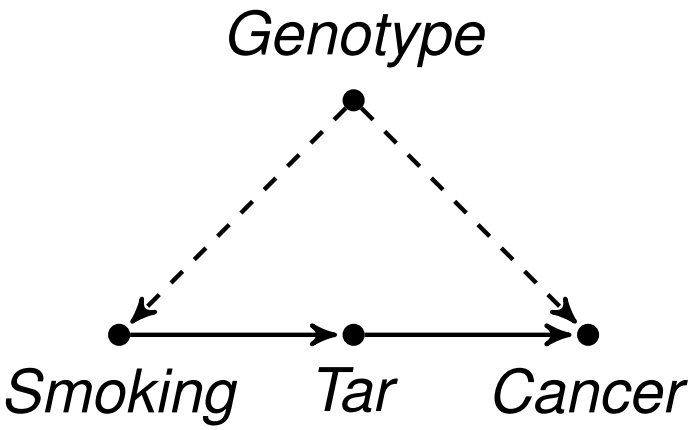
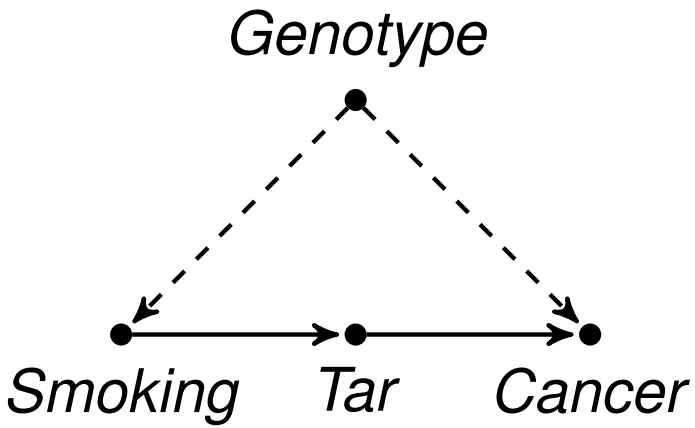


Solved

Shut the front door!



Can you ID the effect of smoking on tar? How?

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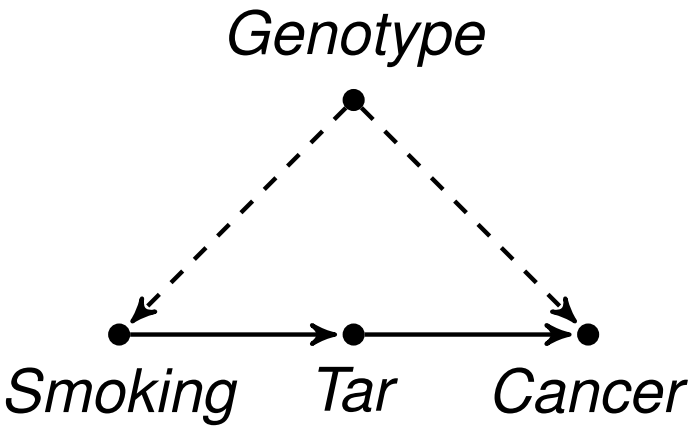

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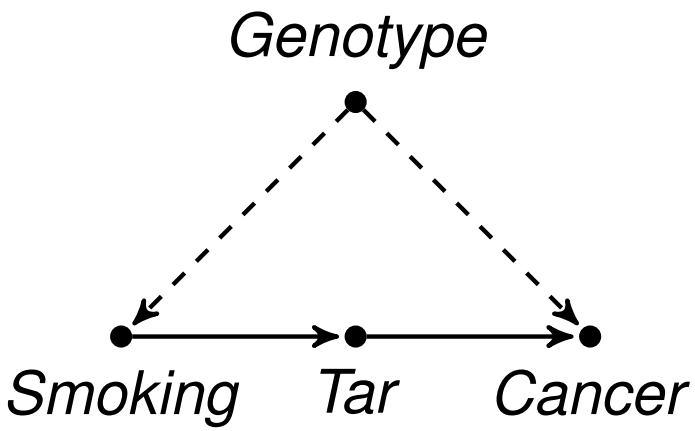

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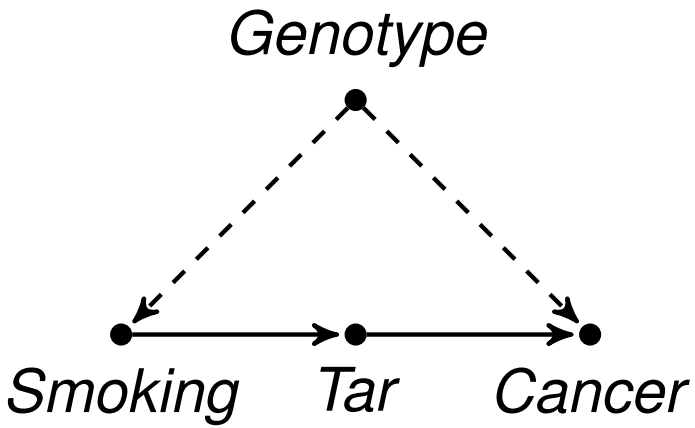

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- And we have those pieces:

$$p(c|do(s)) = \sum_t \sum_{s'} p(C = c|T = t, S = s')p(S = s')p(T = t|S = s)$$

where s' is just an index, to be distinguished from s.

Are those all the ID strategies DAGs show us?

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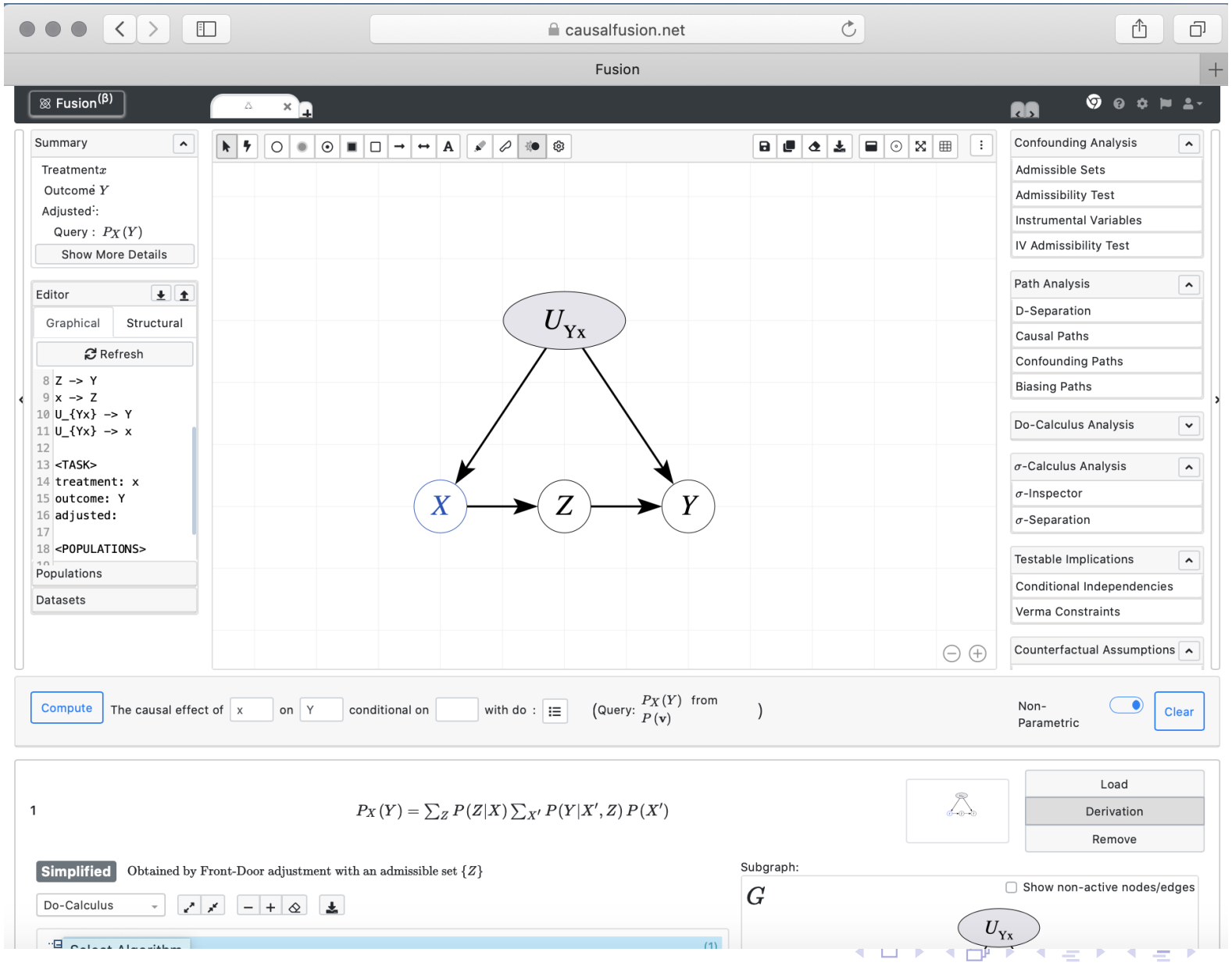
Causal Fusion – give it a try



The identification landscape

Useful to have this picture in mind:

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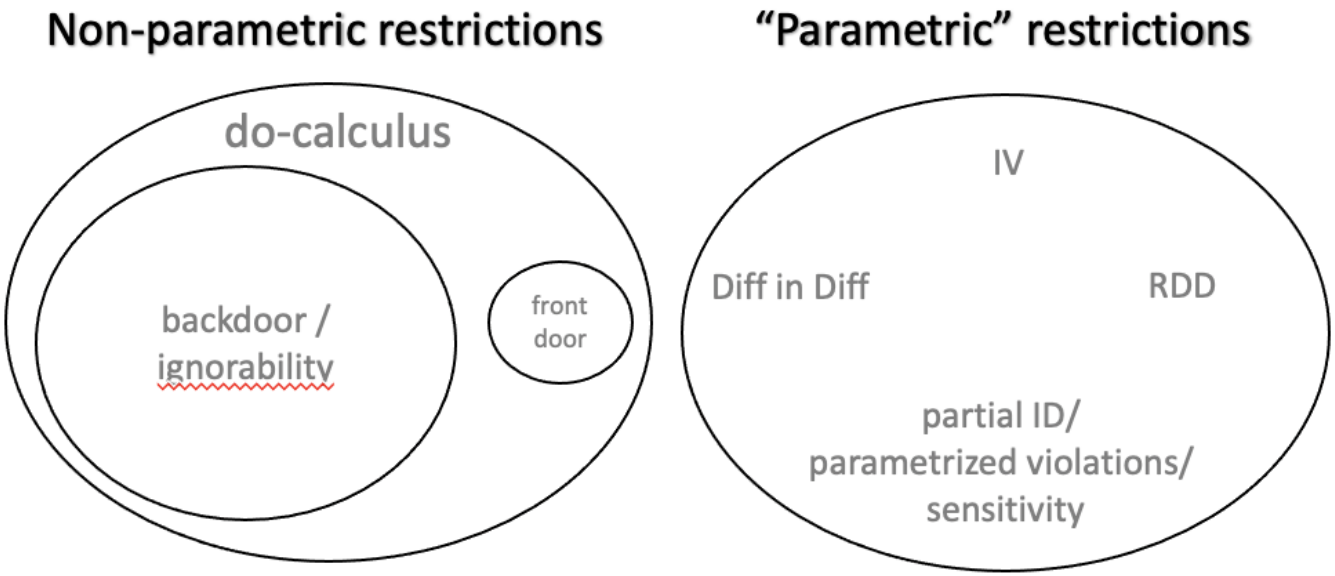
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